
Design Document for **Multi Level Donkey Kong**

Group **2_mstafa_2**

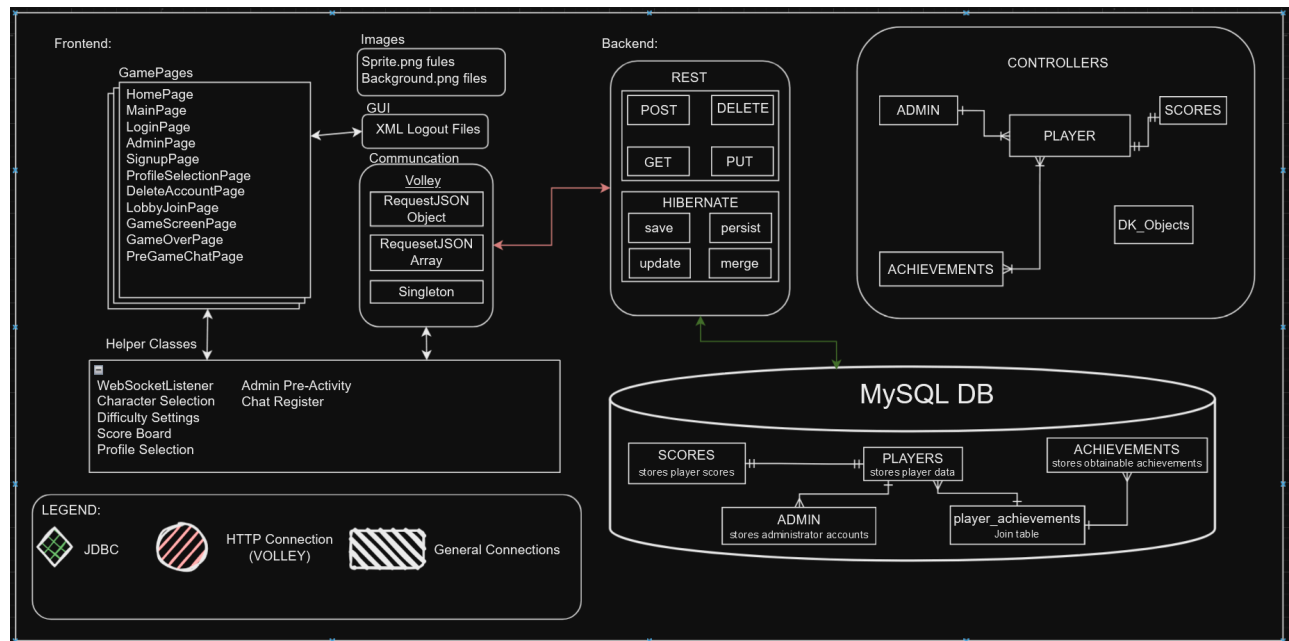
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Block Diagram:



Frontend (Currently Implemented)

Lobby page

- becomes available when you login, and then displays server url, chosen sprite, and functionality to go to Pregame Lobby
- Pregame lobby allows you to send a message to other people before starting the game with the “PLAY” button.

Game screen

- Countdown timer which starts at 120 sec, then counts down. Upon completion, the game ends and is replaced by the Game Over Screen.
- Objects (platforms and items/obstacles) have differing logic for how the player interacts. Platforms are solid, obstacles can collide with players and subtract points. Collectibles can collide with players and increase points.

Login Screen

- Returning Player Login: Can only login with a pre-existing account, if none exist players will be directed to New Player Login.
- New Player Login: Allows the player to make a username/password combo for the server to record. There are basic password rules in place, i.e. 4-8 digits only.
- Administrator Login: Allows the user to see all the players who have made an account and be able to delete or modify other players accounts.
- Account Deletion: Any user/player who knows their username and password can choose to delete their account and data by clicking the delete account button.

Difficulty Settings Screen

- List Objects Screen: Lists all objects that can appear on screen while the game is being played.
- Add Objects Screen: Allows administrators to add a new object that can appear on screen.
- Update Scores Screen: Allows administrators to update the score affector value of any object.
- Remove Objects Screen: Allows administrators to delete any object that can appear on screen if they know the ID number associated with that object.
- Home Button: When clicked returns the user to the home screen.

Character selection Screen

- Select Character Buttons: There are six buttons that show the sprite that the user can select as their character in game. Selections will be saved on the server.

Backend:

Communication

- POST: send information to server db
- PUSH: update information on server
- DELETE: remove information
- GET: retrieve information

Controllers

- PLAYER: Controls all players, has a one-to-one relationship with SCORES, many-to-one relationship with ADMIN, and many-to-many relationship with ACHIEVEMENTS. Has variables such as score, id, name, password...
- DK_Objects: Database that maintains information necessary for objects to function.
- ADMIN: Players that are elevated to admin status are stored here
- SCORES: Holds scores for each player
- ACHIEVEMENTS: Stores achievements that each player can obtain.
- COUNTDOWN: Provides server side functionality for game countdown
- CHAT: Stores and enables chat messages for pregame lobby

TABLE RELATIONSHIPS DIAGRAM:

